INCH-POUND

MS27500H
w/AMENDMENT 2
29 October 2024
SUPERSEDING
MS27500H
w/AMENDMENT 1
15 November 2012

DETAIL SPECIFICATION SHEET

CONNECTORS, PLUG, ELECTRICAL, 90° ELBOW, CRIMP TYPE, BAYONET COUPLING, SERIES II

Inactive for new design after 15 November 2012. For new design, use MS27473 with SAE-AS85049/47.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-38999.

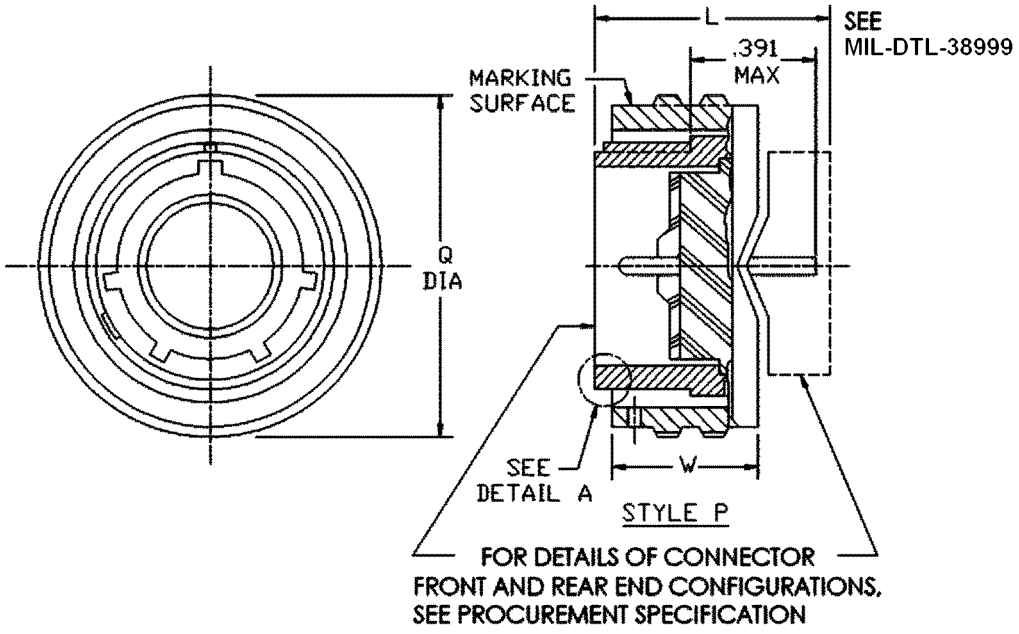
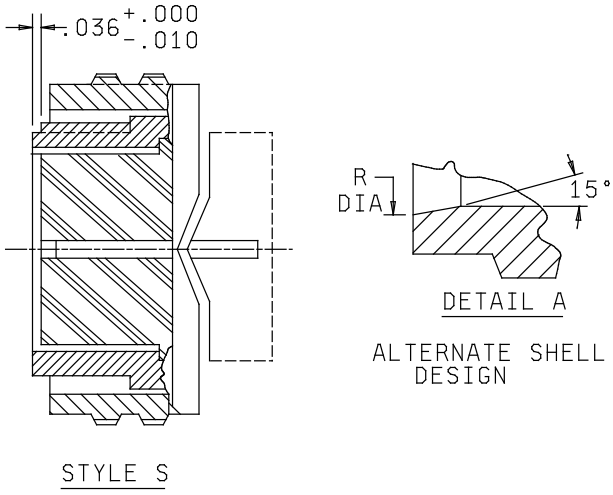


FIGURE 1. Plug, class E.

AMSC N/A FSC 5935

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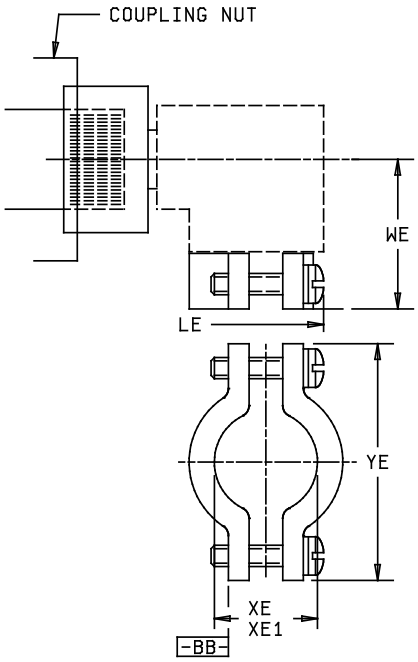


FIGURE 1. Plug, class E – Continued.

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Shell size	Q max dia	R dia +.005 -.006	W ±. 015	LE max overall length	WE max	XE min cable	XE1 max cable	YE max
8	.750	.316	.547	1.859	1.000	.082	.234	.828
10	.859	.436		1.859	1.032	.082	.234	.891
12	1.031	.549		1.938	1.094	.114	.328	1.016
14	1.156	.675		2.125	1.219	.176	.457	1.141
16	1.281	.799		2.281	1.281	.238	.634	1.255
18	1.406	.893		2.312	1.344	.208	.614	1.469
20	1.531	1.018	.524	2.375	1.375	.302	.698	1.550
22	1.641	1.143		2.531	1.438	.302	.823	1.675
24	1.766	1.268	.575	2.828	1.719	.332	.853	1.800

Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.005	0.13	.316	8.03	.750	19.05	1.141	28.98	1.641	41.68
.006	0.15	.328	8.33	.799	20.29	1.143	29.03	1.675	42.55
.010	0.25	.332	8.43	.823	20.90	1.156	29.36	1.719	43.66
.015	0.38	.391	9.93	.828	21.03	1.219	30.96	1.766	44.86
.036	0.91	.436	11.07	.853	21.67	1.255	31.88	1.800	45.72
.082	2.08	.457	11.61	.859	21.82	1.268	32.21	1.859	47.22
.109	2.77	.524	13.31	.891	22.63	1.281	32.54	1.938	49.23
.114	2.90	.547	13.89	.893	22.68	1.344	34.14	2.125	53.98
.155	3.94	.549	13.94	1.000	25.40	1.375	34.93	2.281	57.94
.176	4.47	.575	14.61	1.016	25.81	1.406	35.71	2.312	58.72
.208	5.28	.614	15.60	1.018	25.86	1.438	36.53	2.375	60.33
.234	5.94	.634	16.10	1.031	26.19	1.469	37.31	2.531	64.29
.238	6.05	.675	17.15	1.032	26.21	1.531	38.89	2.828	71.83
.302	7.67	.698	17.73	1.094	27.79	1.550	39.37		

NOTES:

1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. For main key and key positions, see MIL-DTL-38999.
4. Dimensions LE and XE are applicable when the end of the screw is flush with surface BB.

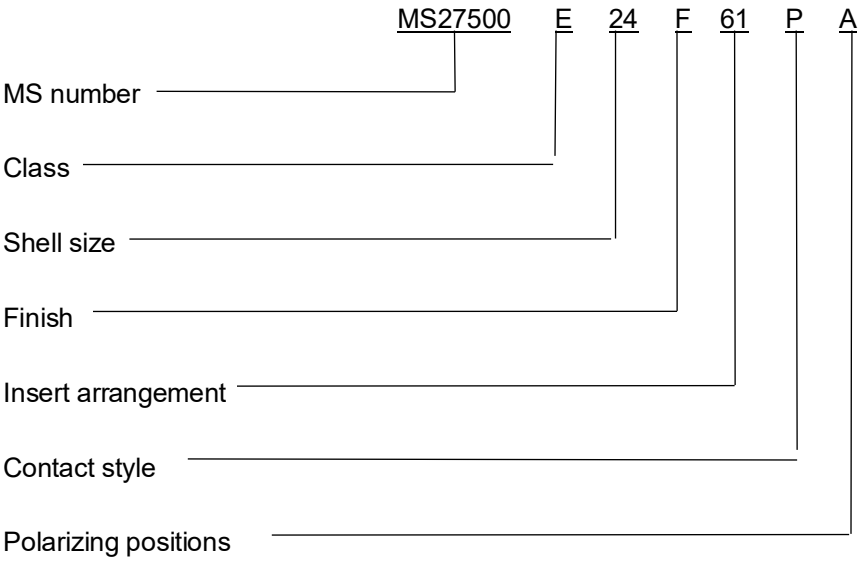
FIGURE 1. Plug, class E – Continued.

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REQUIREMENTS:

Dimensions and configuration: See figure 1.
Interface dimensions shall be in accordance with MIL-DTL-38999.
Mating connectors: MIL-DTL-38999/10, MS27472, MS27474, MS27475, MS27476, MS27477, MS27478, MS27479, MS27480, MS27497, MS27499, MS27508 and MS27513.
Procurement specification: MIL-DTL-38999.
Insert arrangements: See MIL-STD-1560.
For class E, compression nut in accordance with SAE-AS85049/27 shall be included.

Part or Identifying Number (PIN) example:



No letter is required for normal.

Restrictions.
Finish F is not for Navy use. Finish F is inactive for new design for Air Force.

Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Referenced documents. In addition to MIL-DTL-38999, this document references the following:

MIL-DTL-38999/10
MIL-STD-1560
MS27472
MS27473
MS27474
MS27475
MS27476

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MS27477
MS27478
MS27479
MS27480
MS27497
MS27499
MS27508
MS27513
SAE-AS85049/27
SAE-AS85049/47

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – AS
Air Force – 85
DLA – CC

Preparing activity:
DLA – CC

(Project 5935-2024-171)

Review activities:
Army – AR, MI
Navy – EC, MC, OS
Air Force – 19

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